

Module 2: EBS Assignment

Problem Statement:

You work for XYZ Corporation. Your corporation wants to launch a new web-based application using AWS Virtual Machines. Configure the resources accordingly with appropriate storage for the tasks.

Tasks Performed:

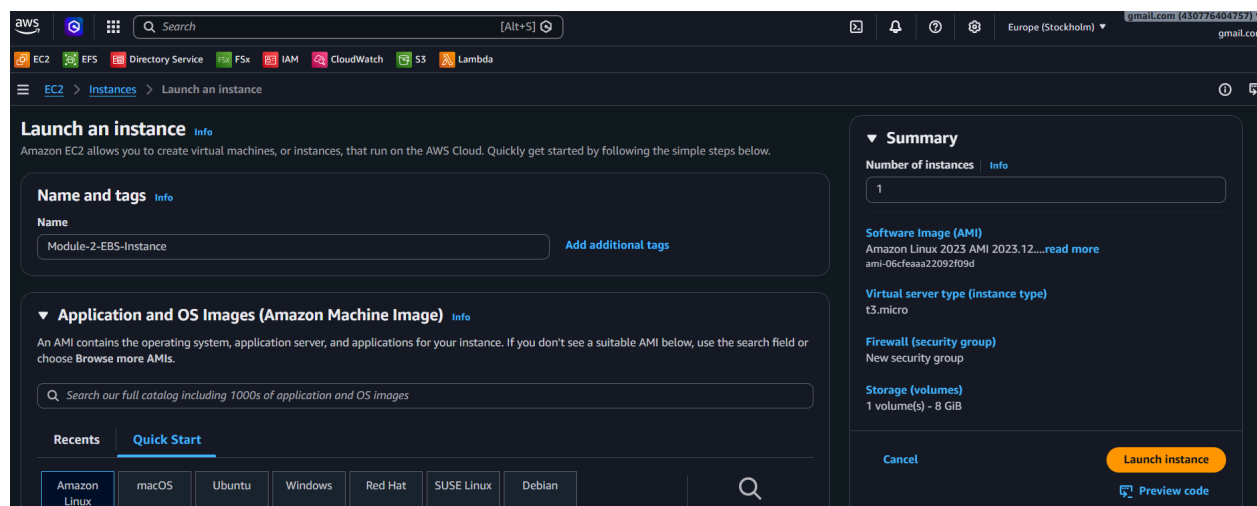
1. Launched a Linux EC2 instance.
2. Created an EBS volume with 20 GB of storage and attached it to the created EC2 instance.
3. Resized the attached volume and made sure it reflects in the connected instance.

Task 1 — Launched Linux EC2

EC2 → Instances → Launch instances

Configured:

- Name: Module-2-EBS-Instance
- AMI: Amazon Linux 2023
- Instance type: t3.micro
- Selected a key pair.



Launched the instance.

Instance summary for i-02e4ed84afe923a74 (Module-2-EBS-Instance) Info

Updated less than a minute ago

[Refresh](#) [Connect](#) [Instance state](#) [Actions](#)

Instance ID i-02e4ed84afe923a74	Public IPv4 address 13.60.78.54 open address	Private IPv4 addresses 172.31.33.218
IPv6 address -	Instance state Running	Public DNS ec2-13-60-78-54.eu-north-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-33-218.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-33-218.eu-north-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 13.60.78.54 [Public IP]	VPC ID vpc-0e9d79ad3a9956ee4	Auto Scaling Group name -
IAM role -	Subnet ID subnet-01684b45c06381071	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:eu-north-1:430776404757:instance/i-02e4ed84afe923a74	

Task 2 — Create and Attach the 20 GB EBS Volume

1. Create the EBS volume

AWS Console → EC2 → Elastic Block Store → Volumes → Create volume.

Selected:

Volume Type : gp2

Size : 20GiB

Create volume Info

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type Info
General Purpose SSD (gp3)

Size (GiB) Info
20
Min: 1 GiB, Max: 65536 GiB.

IOPS Info
3000
Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) Info
125
Min: 125 MiB, Max: 2000 MiB, Baseline: 125 MiB/s.

Availability Zone Info
eu-north-1b

Availability Zone : Same AZ as the EC2 instance

Availability Zone [Info](#)

eun1-az2 (eu-north-1b) ▼

2. Attach it to your EC2 instance

Selected the newly created volume.

Actions → **Attach volume**

Volumes (1/2) [Info](#)

Last updated less than a minute ago [Refresh](#) [Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets [Choose filter set](#)

	Name	Volume ID	Type	Size	IOPS	Throughput	Owner ID	Snapshot ID	
☒		vol-00e6ff964b16198e4	gp3	20 GiB	3000	125	430776404757	-	Modify volume Create snapshot Create snapshot lifecycle policy Delete volume Attach volume
☐		vol-09428655bf7e51875	gp3	8 GiB	3000	125	430776404757	snap-07c595a...	

Selected the EC2 instance : Module-2-EBS-Instance

Selected a device name : /dev/sde

Clicked **Attach volume**.

Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID
 vol-00e6ff964b16198e4

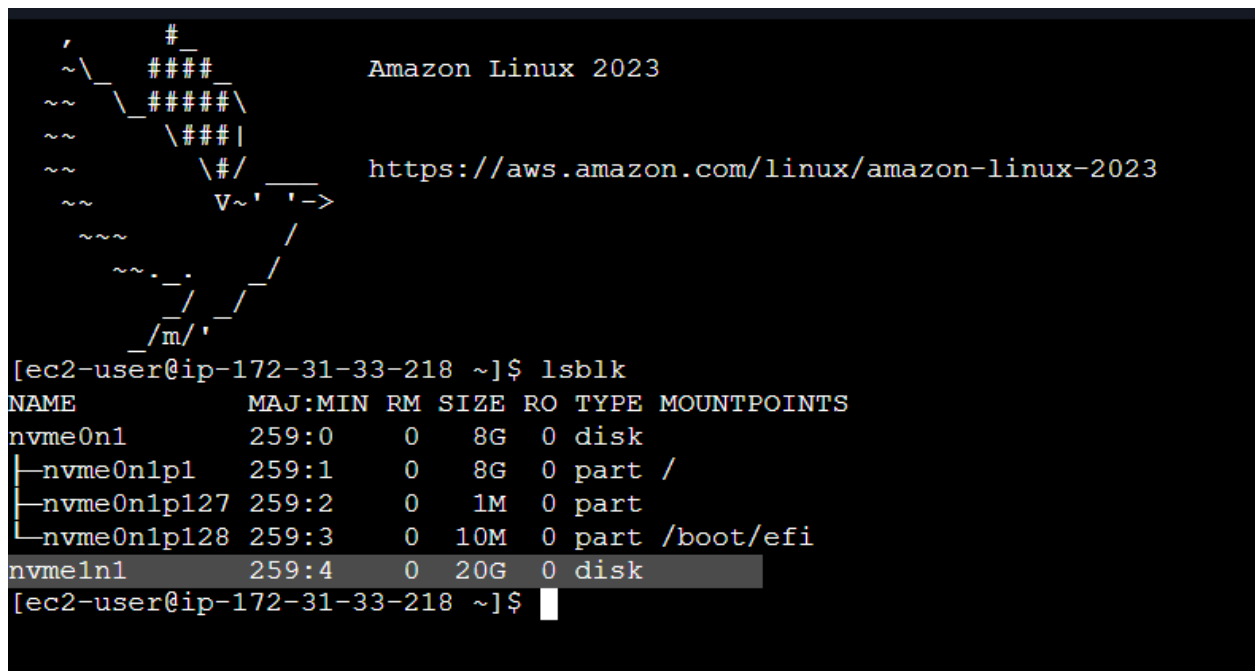
Availability Zone
eun1-az2 (eu-north-1b)

Instance [Info](#)
i-02e4ed84afe923a74
(Module-2-EBS-Instance) (running) [Refresh](#)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)
/dev/sde ▼

Recommended device names for Linux: /dev/xvda for root volume. /dev/sd[f-p] for data volumes.



3. Created a mount directory and Mounted the EBS volume

Created the mount point:

```
[ec2-user@ip-172-31-33-218 ~]$ sudo mkdir /data
```

Mounted the EBS volume:

```
[ec2-user@ip-172-31-33-218 ~]$ sudo mount /dev/nvme0n1p128 /data
```

Verified

```
[ec2-user@ip-172-31-33-218 ~]$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1                            259:0    0   8G  0 disk
├─nvme0n1p1                        259:1    0   8G  0 part /
├─nvme0n1p127                     259:2    0   1M  0 part
└─nvme0n1p128                     259:3    0  10M  0 part /data
                                                /boot/efi
nvme1n1                            259:4    0  20G  0 disk
[ec2-user@ip-172-31-33-218 ~]$
```

Task 3 — Resize EBS from 20 GiB → 30 GiB

EC2 → Volumes

1. Selected the EBS volume.
2. **Actions** → **Modify volume**.
3. Change: Size: 20 → 30 GiB
4. Clicked **Modify**.

The screenshot shows the 'Modify volume' page in the AWS Management Console. The breadcrumb navigation at the top reads: EC2 > Volumes > vol-00e6ff964b16198e4 > Modify volume. The main section is titled 'Volume details'. It contains four sections, each with a label and an 'Info' link: 'Volume ID' with the value 'vol-00e6ff964b16198e4', 'Volume type' with the value 'General Purpose SSD (gp3)', 'Size (GiB)' with a text input field containing '30' and a range 'Min: 1 GiB, Max: 65536 GiB.', and 'IOPS' with a text input field containing '3000' and a range 'Min: 3000 IOPS, Max: 80000 IOPS.'. Below these is the 'Throughput (MiB/s)' section with a text input field containing '125' and a range 'Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.'.

Confirm the modification.

The screenshot shows the 'Volumes' list page in the AWS Management Console. The page title is 'Volumes (2) Info'. There are buttons for 'Recycle Bin', 'Actions', and 'Create volume'. A table lists the volumes with columns: Name, Volume ID, Type, Size, IOPS, Throughput, Owner ID, Snapshot ID, Source volume ID, and Created. The table contains two rows of data.

	Name	Volume ID	Type	Size	IOPS	Throughput	Owner ID	Snapshot ID	Source volume ID	Created
☐		vol-00e6ff964b16198e4	gp3	30 GiB	3000	125	430776404757	-	-	2026/09/26 21:49 GMT+5:...
☐		vol-09428655bf7e51875	gp3	8 GiB	3000	125	430776404757	snap-07c595a...	-	2026/09/26 21:43 GMT+5:...

Verified on EC2:

```
[ec2-user@ip-172-31-33-218 ~]$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1                             259:0    0   8G  0 disk
├─nvme0n1p1                         259:1    0   8G  0 part /
├─nvme0n1p127                       259:2    0   1M  0 part
└─nvme0n1p128                       259:3    0  10M  0 part /data
                                      /boot/efi
nvme1n1                             259:4    0  30G  0 disk
[ec2-user@ip-172-31-33-218 ~]$
```

i-02e4ed84afe923a74 (Module-2-EBS-Instance)